

Mithi Jain

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EDUCATION

VIT Bhopal University — School of Computer Science Engineering

B.Tech in Computer Science and Engineering, CGPA: 8.23/10.0

2023 – 2027

St. Joseph's Convent Sr. Sec. School

10th Standard: 94.6%

2021

TECHNICAL SKILLS

Languages: Java, Python, SQL

Frameworks & Libraries: Spring Boot, Spring Security, Hibernate, React, PyTorch, Pandas, NumPy

Cloud & DevOps: AWS, Terraform (IaC), Docker, GitHub Actions (CI/CD), Apache Kafka, Prometheus, Grafana, OpenTelemetry

Databases & Tools: PostgreSQL, MySQL, Git, Maven, JUnit

CS Fundamentals: Data Structures & Algorithms (DSA), Design & Analysis of Algorithms (DAA), OOP, OS, DBMS, Computer Networks (CN), Distributed Systems, System Design

EXPERIENCE

Junior Developer Intern — Apiero Medica Pvt. Ltd. (MyAarogya)

Dec 2025 – Mar 2026

- Contributed to backend development of a healthcare platform connecting rural patients with urban doctors via real-time video consultations, and developed CRUD-based REST APIs for prescription data.
- Collaborated with senior developers on scalable healthcare system architecture, backend integration, and live medical data handling in an Agile team environment.

Tech Stack: Java, Spring Boot, REST APIs, MySQL, PostgreSQL, Hibernate, Maven, Git

PROJECTS

Sentinel — Real-Time Fraud Detection & Response Platform

Apr – Jun 2026

- Engineered an event-driven microservices pipeline using Apache Kafka to compute real-time features with point-in-time correctness, sustaining ~1,200 events/sec (tested to 50,000-event replay runs) at a 0% feature-mismatch rate, down from 6.4% in the naive/leakage version.
- Achieved 94% fraud-burst detection accuracy with a 1.8% false-positive rate at 180ms (p50) / 640ms (p99) ingest-to-decision latency; orchestrated a saga pattern across services achieving a 99.2% success rate with ~8.4s automatic recovery after a mid-saga service failure.
- Deployed on AWS via Terraform (IaC) with CI/CD (GitHub Actions) and 100% OpenTelemetry tracing coverage; validated via shadow evaluation at 91.3% live/shadow rule agreement and zero duplicate effects across repeated-event chaos tests.

Tech Stack: Java, Spring Boot, Apache Kafka, PostgreSQL, AWS, Terraform, Docker, GitHub Actions, OpenTelemetry, Prometheus, Grafana, JUnit

Chronos — Distributed Job Scheduler (Raft Consensus, from scratch)

Feb – Mar 2026

- Implemented Raft consensus (leader election, log replication, commit safety, write-ahead persistent log) from scratch in pure Java 17+ with zero external dependencies, using records and sealed interfaces for RPC message types and node states, a hand-rolled JSON parser, and JDK HttpServer/HttpClient transport across a 6-module architecture.
- Verified correctness with a custom test-runner suite and a real 5-node, multi-process chaos harness that kills real OS processes and simulates network partitions, passing 13/13 checks across multiple runs with zero committed-job loss and zero duplicate execution; manually re-verified via a live dashboard with real process kill/restart and partition/heal controls.
- Designed fencing-token idempotency (per-job version numbers) so stale worker reports after reassignment are silently ignored; caught and fixed 4 real concurrency/timing bugs during development, including a stale-leader-reconnect bug and cross-OS fsync timeout tuning.

Tech Stack: Java 17 (records, sealed interfaces), Multithreading, JDK HttpServer/HttpClient, Custom JSON Parser, Raft Consensus, Distributed Systems, Fault Tolerance

Intelligent Java Cache

Oct – Nov 2025

- Designed a multithreaded, concurrent in-memory caching library from scratch (LRU, LFU, ARC, TinyLFU) using HashMap + doubly linked list core data structures, ReentrantLock-based thread safety, and the Decorator design pattern; sustained 4M+ ops/sec under 16-thread load with zero data corruption across 640K+ operations.
- Built TinyLFU's Count-Min Sketch admission filter, achieving a 100% vs. 0% hot-key survival rate against LRU under scan floods, and a 78.2% vs. 71.8% hit rate (LFU vs. LRU) on Zipfian access patterns.
- Created an adaptive layer that shadow-evaluates all four eviction policies live and auto-switches to the best performer within 1,000 operations; validated with 60+ automated JUnit tests across a 3-module REST/WebSocket architecture.

Tech Stack: Java 17, HashMap, Doubly Linked List, ReentrantLock, ConcurrentHashMap, Count-Min Sketch, Spring Boot, WebSocket, React, JUnit, Maven, Docker

NovelTea — Book Discovery & Recommendation Platform

Aug 2025

- Built a full-stack app (Java, Spring Boot, React) with JWT auth (Spring Security, BCrypt), a PostgreSQL/JPA relational data model (DTO pattern), and Google Books + Gemini LLM API integration for AI-grounded recommendations.
- Cut cached API latency from ~300–800ms to ~1ms using Caffeine caching; containerized with Docker, deployed via Render/Vercel with GitHub Actions CI.

Tech Stack: Java, Spring Boot, Spring Security, BCrypt, JPA/Hibernate, PostgreSQL, React, Docker, GitHub Actions, Google Books & Gemini APIs

CERTIFICATIONS

Mastering Data Structures & Algorithms (FACE Prep) • DP-900: Microsoft Azure Data Fundamentals • The Bits and Bytes of Computer Networking (Google, Coursera) • Introduction to Machine Learning (NPTEL)